

M1 LESSON 1

# Project Management



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# Learning Outcomes

At the end of this lesson, the student is expected to be able to:

- Explain what a project is, provide examples of IT Projects, list various attributes of projects, and describe the tripe constraint of project management.
- Describe project management and discuss key elements of the project management framework, including project stakeholders, the project management knowledge areas, common tools and techniques, and project success.
- Discuss the relationship between project, program, and portfolio management and the contributions each makes to enterprise success.
- Understand the role of project managers by describing what they do, what skills they need, and career opportunities for IT project managers.
- Describe the project management profession, including its history, the role of professional organizations like the Project Management Institute (PMI), the importance of certification and ethics, and the advancement of project management software.

**Project**

**Operation**

# Project

temporary endeavor undertaken to  
create a unique product, service, or  
result

# Operation



# Project

temporary endeavor undertaken to create a unique product, service, or result

# Operation

work done in organizations to sustain the business.

**temporary**

**unique purpose**

**developed using**

**progressive**

**elaboration**

# **Project**

**uncertainty**

attributes

**requires resources**

**primary customer or**

**sponsor**

**APPLICATION  
DEVELOPMENT**

**IS UPDATES**

**SYSTEM  
DEVELOPMENT**

**IT**

**PROJECTS**

# PROJECT CONSTRAINTS

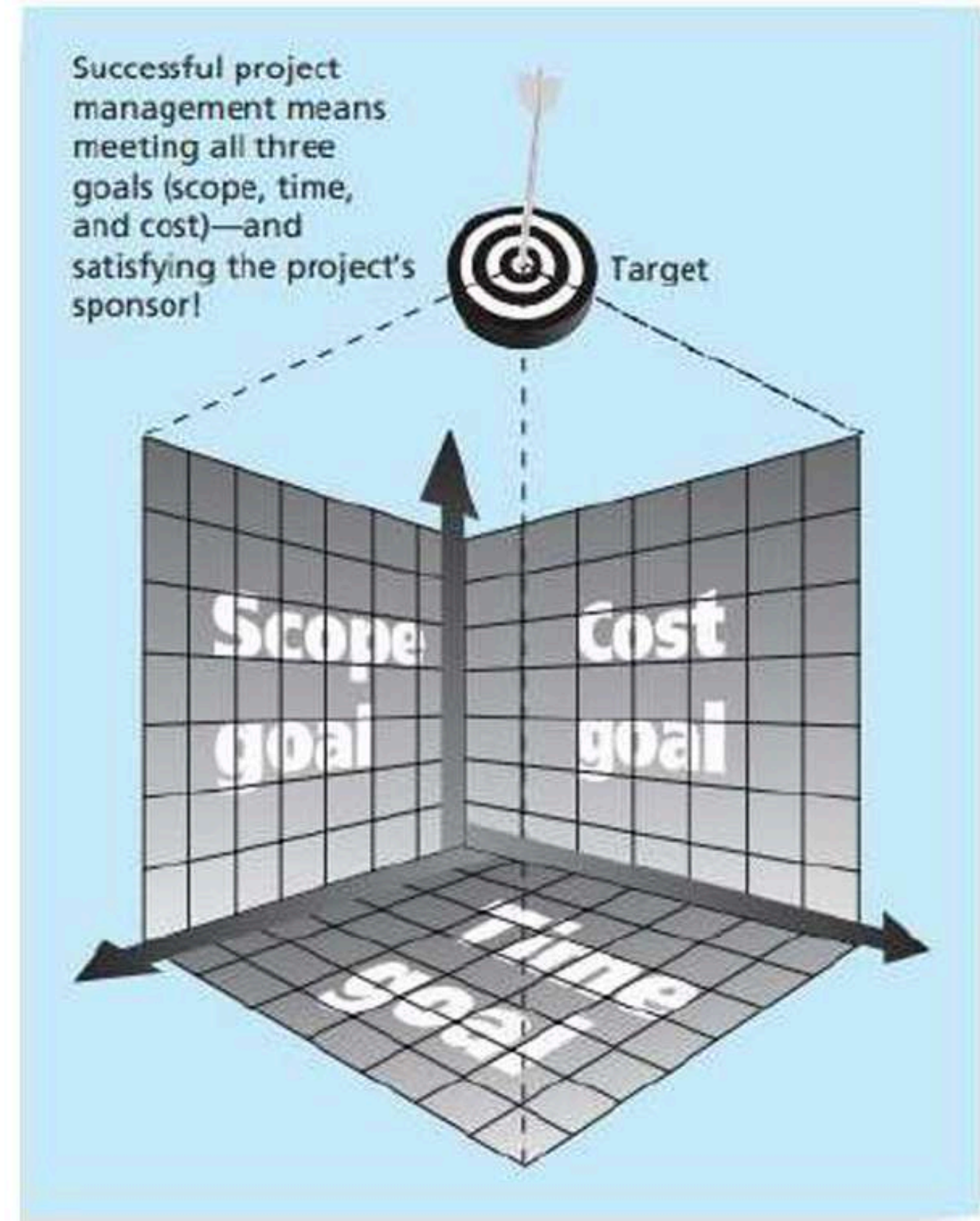
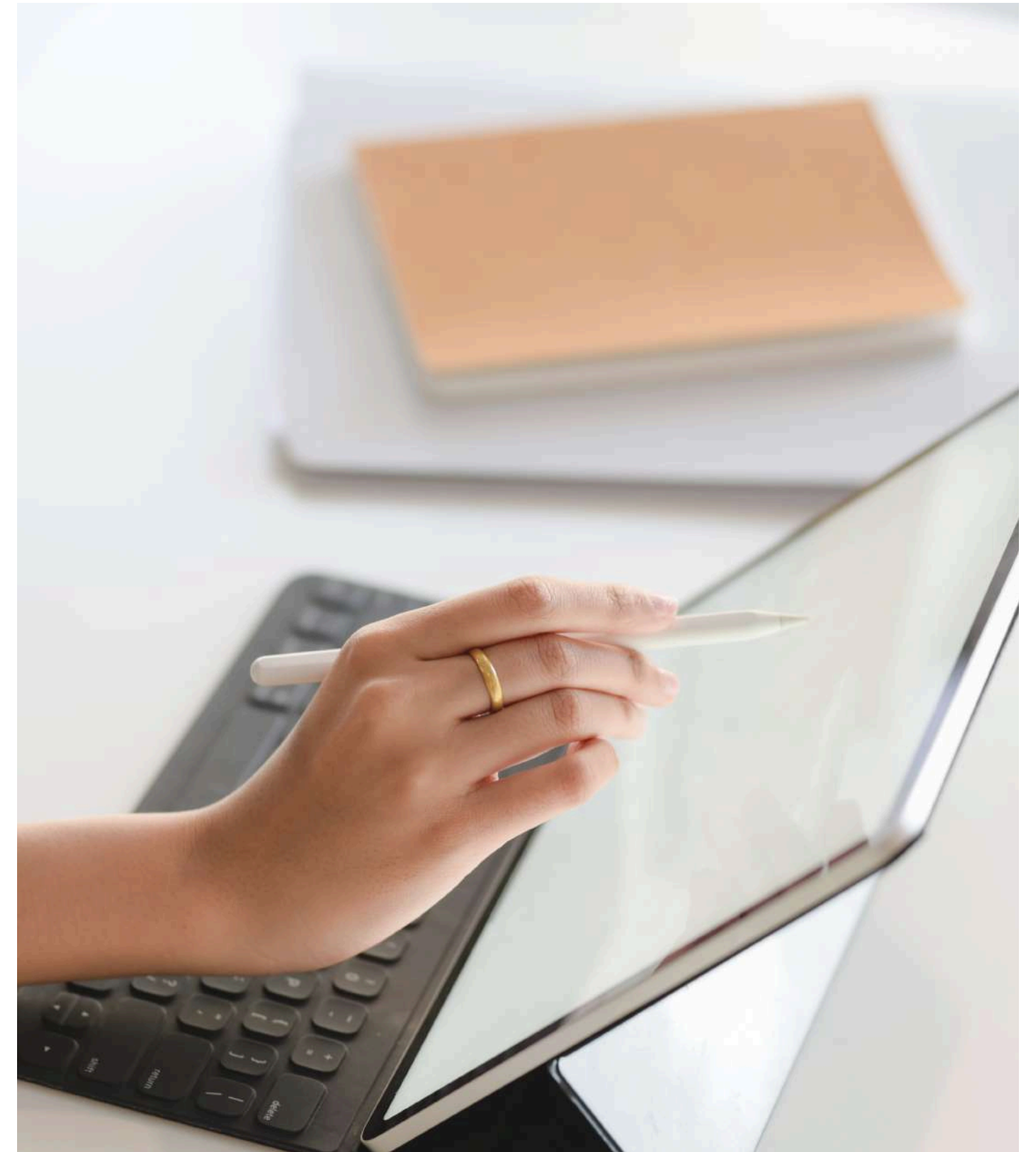


FIGURE 1.1 PROJECT CONSTRAINTS



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# **Project Management?**



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# **Project Management**

the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.

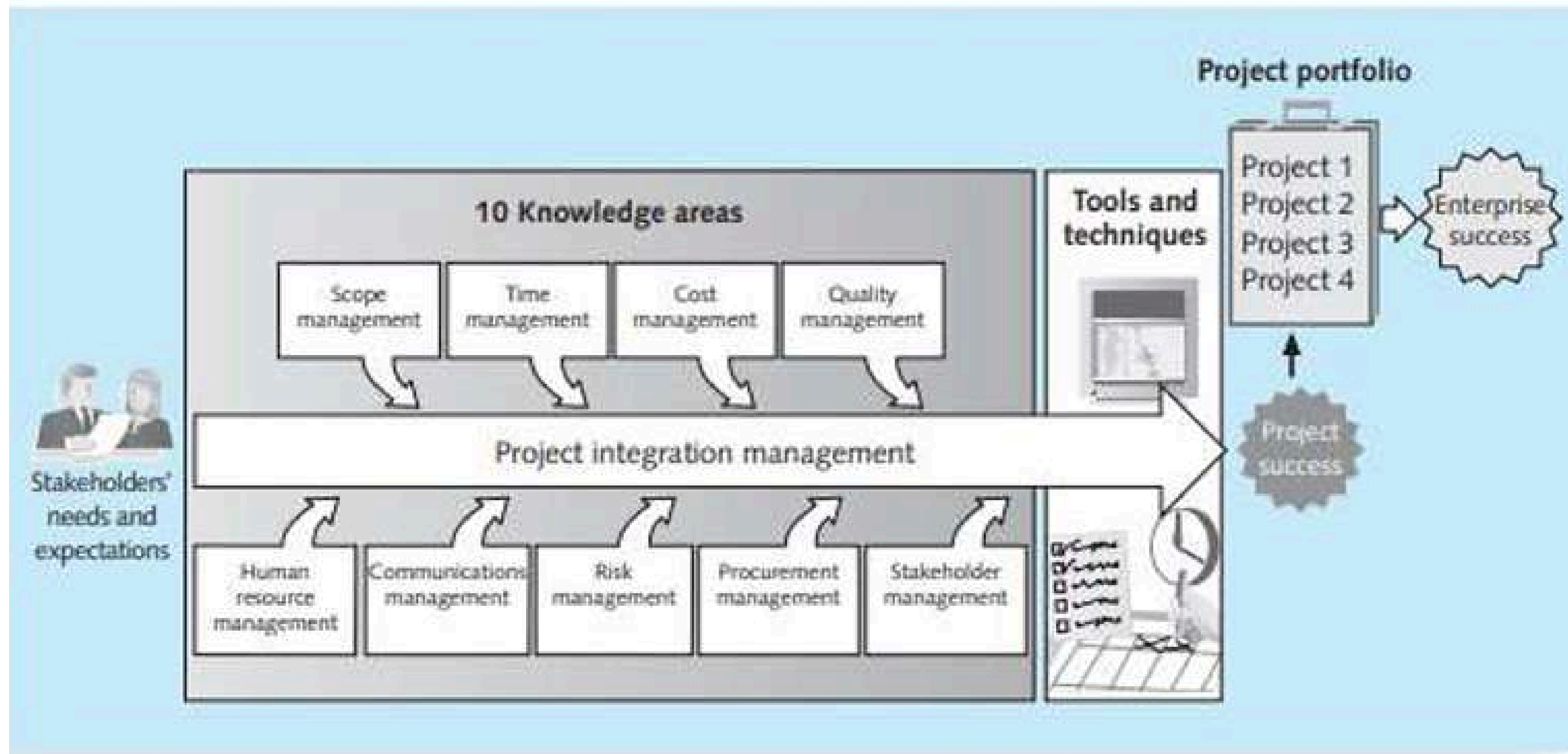


FIGURE 1.2 PROJECT MANAGEMENT FRAMEWORK



**project sponsor**

**financial institution**

**contractor**

**constructions workers,  
electricians, carpenters**

**support staff**

**suppliers**

**project opponents**

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# **Project Stakeholders**

people involved in or affected by project activities, and include the project sponsor, project team, support staff, customers, users, suppliers, and even opponents of the project.



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Why is it important to  
determine the project  
stakeholders?

**TABLE 1-1** Common project management tools and techniques by knowledge area

| Knowledge Area/Category | Tools and Techniques                                                                                                                                                                                       | Super Tools                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Integration management  | Project selection methods<br>Project management methodologies<br>Stakeholder analyses<br>Work requests<br>Project charters<br>Project management plans<br>Change control boards<br>Project review meetings | Project management software<br>Change requests<br>Lessons-learned reports |
| Scope management        | Statements of work<br>Scope management plans<br>Scope verification techniques<br>Scope change controls                                                                                                     | Scope statements<br>Work breakdown structures<br>Requirements analyses    |
| Time management         | Project network diagrams<br>Critical path analysis<br>Crashing<br>Fast tracking<br>Schedule performance measurements                                                                                       | Gantt charts                                                              |
| Cost management         | Project budgets<br>Net present value<br>Return on investment<br>Payback analysis<br>Earned value management<br>Project portfolio management<br>Cost estimates<br>Cost management plans<br>Cost baselines   |                                                                           |

| Quality management        | Quality metrics<br>Checklists<br>Quality control charts<br>Pareto diagrams<br>Fishbone diagrams<br>Maturity models<br>Statistical methods<br>Test plans               |                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Human resource management | Motivation techniques<br>Empathic listening<br>Responsibility assignment matrices<br>Project organizational charts<br>Resource histograms<br>Team building exercises  |                                       |
| Knowledge Area/Category   | Tools and Techniques                                                                                                                                                  | Super Tools                           |
| Communications management | Communications management plans<br>Conflict management<br>Communications media selection<br>Status reports<br>Virtual communications<br>Templates<br>Project websites | Kick-off meetings<br>Progress reports |
| Risk management           | Risk management plans<br>Risk registers<br>Probability/impact matrices<br>Risk rankings                                                                               |                                       |
| Procurement management    | Make-or-buy analyses<br>Contracts<br>Requests for proposals or quotes<br>Source selections<br>Supplier evaluation matrices                                            |                                       |

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How to you define  
success or failure of a  
project?



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1. The project met scope, time, and cost goals.
  2. The project satisfied the customer/sponsor
  3. Savings or ROI, making sponsor happy

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Why do some projects  
succeed and others fail?



**TABLE 1-2** What helps projects succeed?

|                                 |
|---------------------------------|
| 1. Executive support            |
| 2. User involvement             |
| 3. Clear business objectives    |
| 4. Emotional maturity           |
| 5. Optimizing scope             |
| 6. Agile process                |
| 7. Project management expertise |
| 8. Skilled resources            |
| 9. Execution                    |
| 10. Tools and infrastructure    |

Source: The Standish Group, “CHAOS Manifesto 2013: Think Big, Act Small” (2013).

# Programs and Portfolio



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# PROGRAM

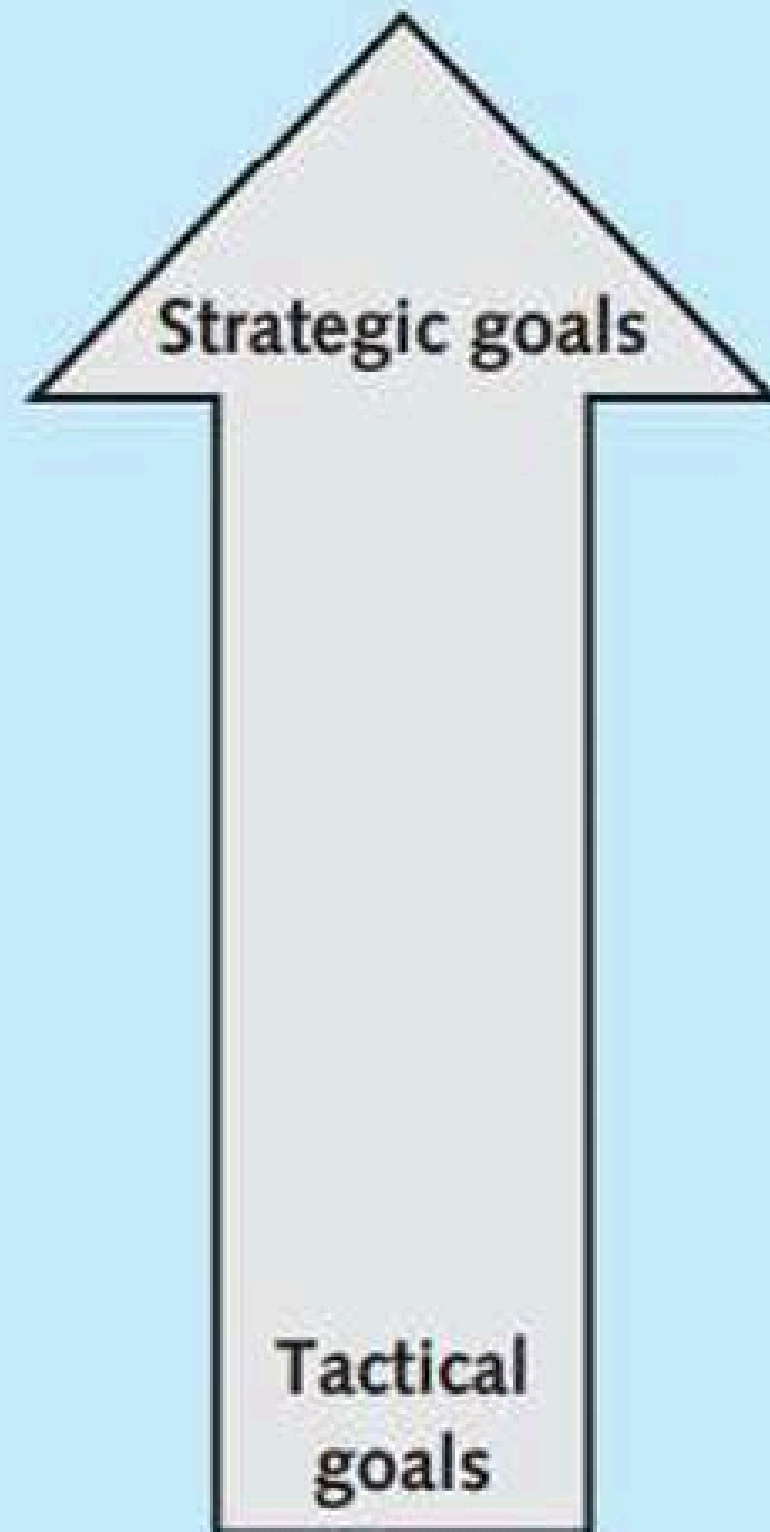
a group of related projects, subprograms, and program activities managed in a coordinated way to obtain benefits and control not available from managing them individually.





# Project Portfolio Management

process of planning, scheduling and monitoring of project portfolios, a process that's typically carried out by project portfolio managers or a project management office (PMO).



### **Project portfolio management**

- Are we working on the right projects?
- Are we investing in the right areas?
- Do we have the right resources to be competitive?

### **Project management**

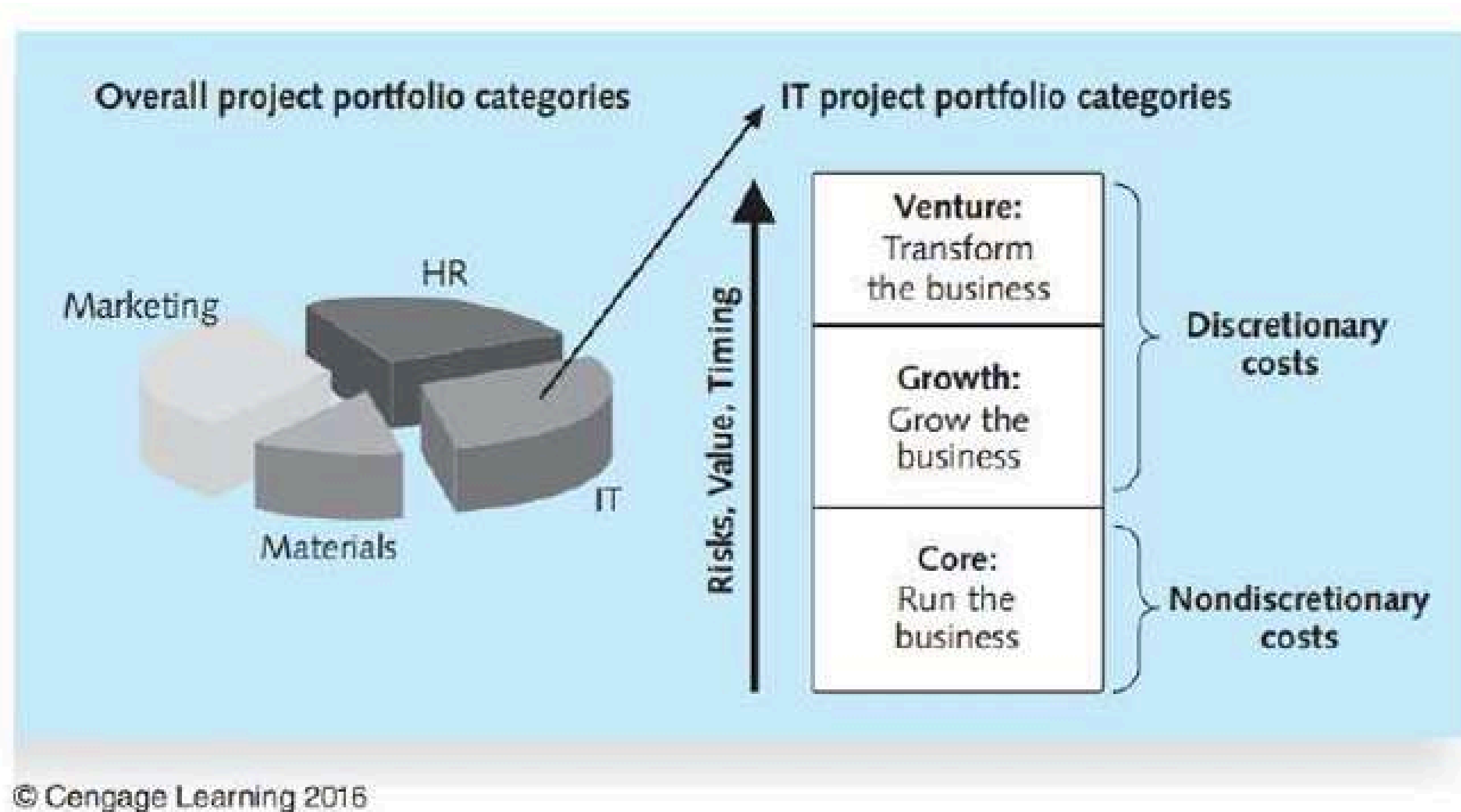
- Are we carrying out projects well?
- Are projects on time and on budget?
- Do project stakeholders know what they should be doing?



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# IT project portfolio categories

1. **Venture:** Projects in this category help transform the business.
2. **Growth:** Projects in this category would help the company increase its revenues.
3. **Core:** Projects in this category must be accomplished to run the business.



**FIGURE 1.4 SAMPLE PROJECT PORTFOLIO APPROACH**

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# **Role of Project Manager**

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# Project Manager Expertise

- The Project Management Body of Knowledge
- Application area knowledge, standards, and regulations
- Project environment knowledge
- General management knowledge and skills
- Soft skills or human relations skills





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# PM needed skills to be effective

- Be highly organized.
- Take charge and know how to lead.
- Be an effective communicator.
- Know how and when to negotiate.
- Be detail-oriented.
- Recognize and solve problems quickly.
- Possess the necessary technical skills

**TABLE 1-3** Ten most important skills and competencies for project managers

|                                               |
|-----------------------------------------------|
| 1. People skills                              |
| 2. Leadership                                 |
| 3. Listening                                  |
| 4. Integrity, ethical behavior, consistency   |
| 5. Strength at building trust                 |
| 6. Verbal communication                       |
| 7. Strength at building teams                 |
| 8. Conflict resolution, conflict management   |
| 9. Critical thinking, problem solving         |
| 10. Understanding and balancing of priorities |

Source: Jennifer Krahn, "Effective Project Leadership: A Combination of Project Manager Skills and Competencies in Context," *PMI Research Conference Proceedings* (July 2006).

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# Based on project situations

- **Large projects:** Leadership, relevant experience, planning, people skills, verbal communication, and team-building skills were most important.
- **High-uncertainty projects:** Risk management, expectation management, leadership, people skills, and planning skills were most important.
- **Innovative projects:** Leadership, people skills, vision- and goal-setting, self-confidence, expectations management, and listening skills were most important.



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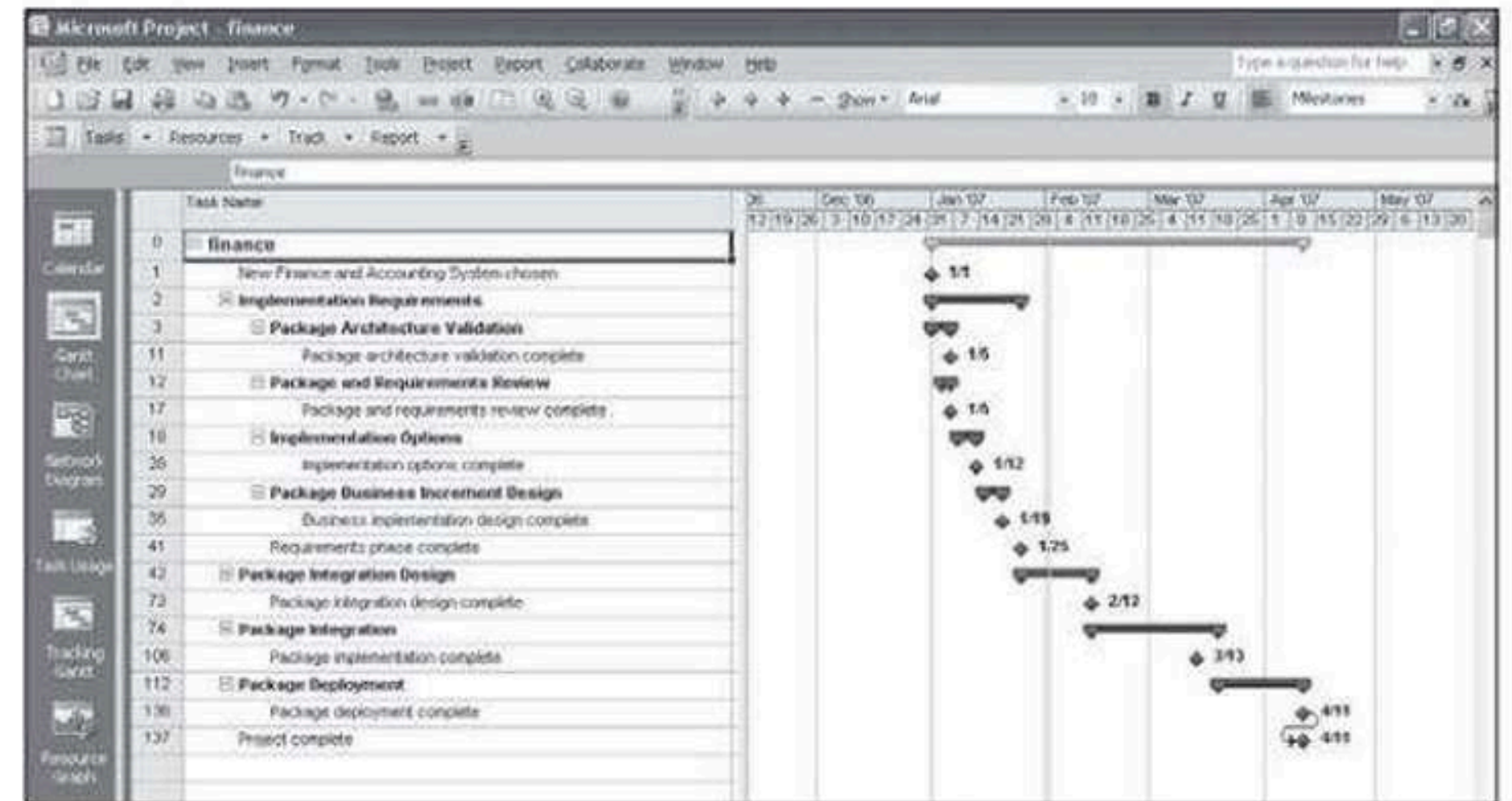
# History of Project Management

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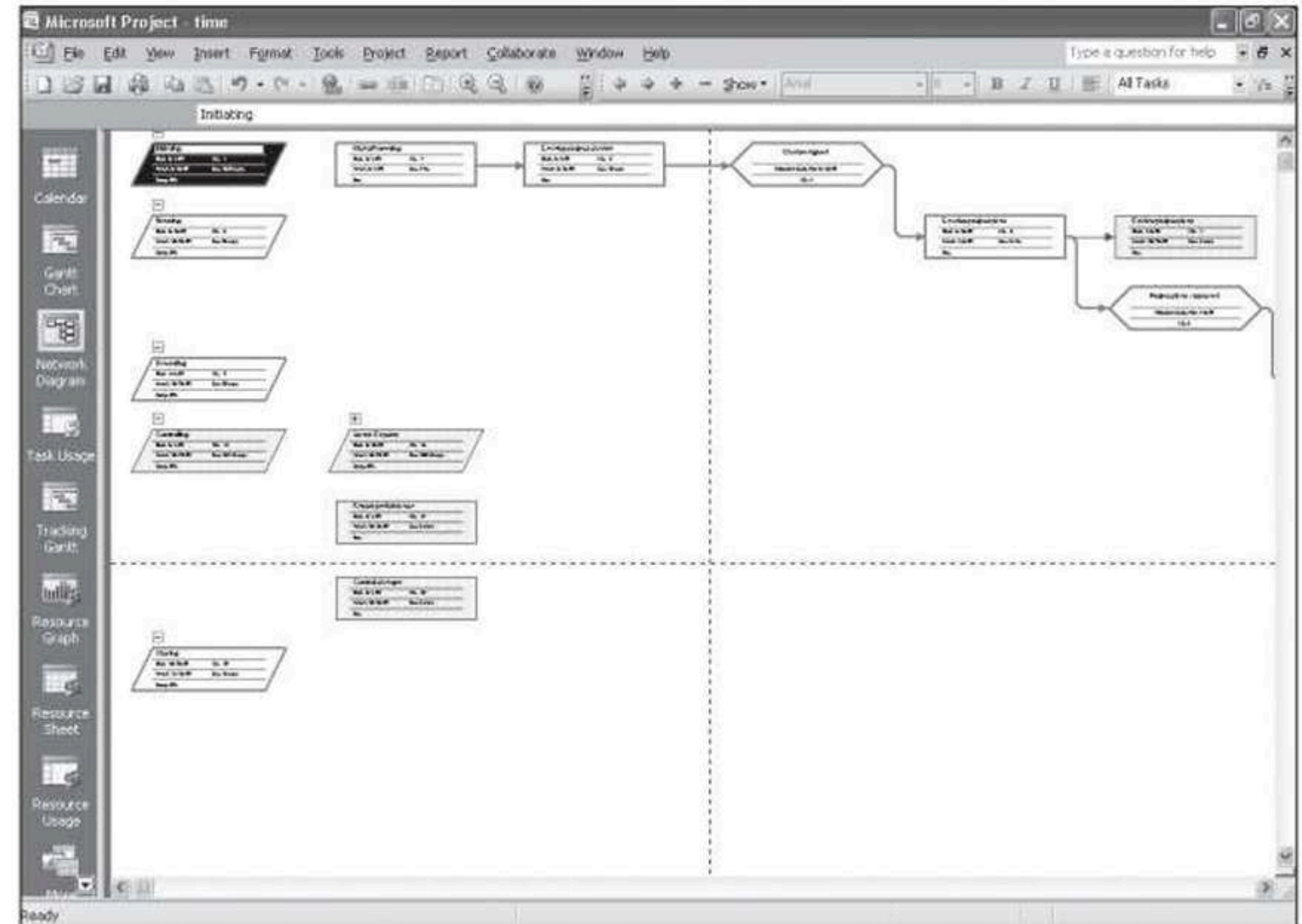
- Henry Gantt (1917)
- displaying project schedule information by listing project activities and their corresponding start and finish dates in calendar form



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FIGURE 1.6 SAMPLE GANTT CHART CREATED WITH PROJECT 13

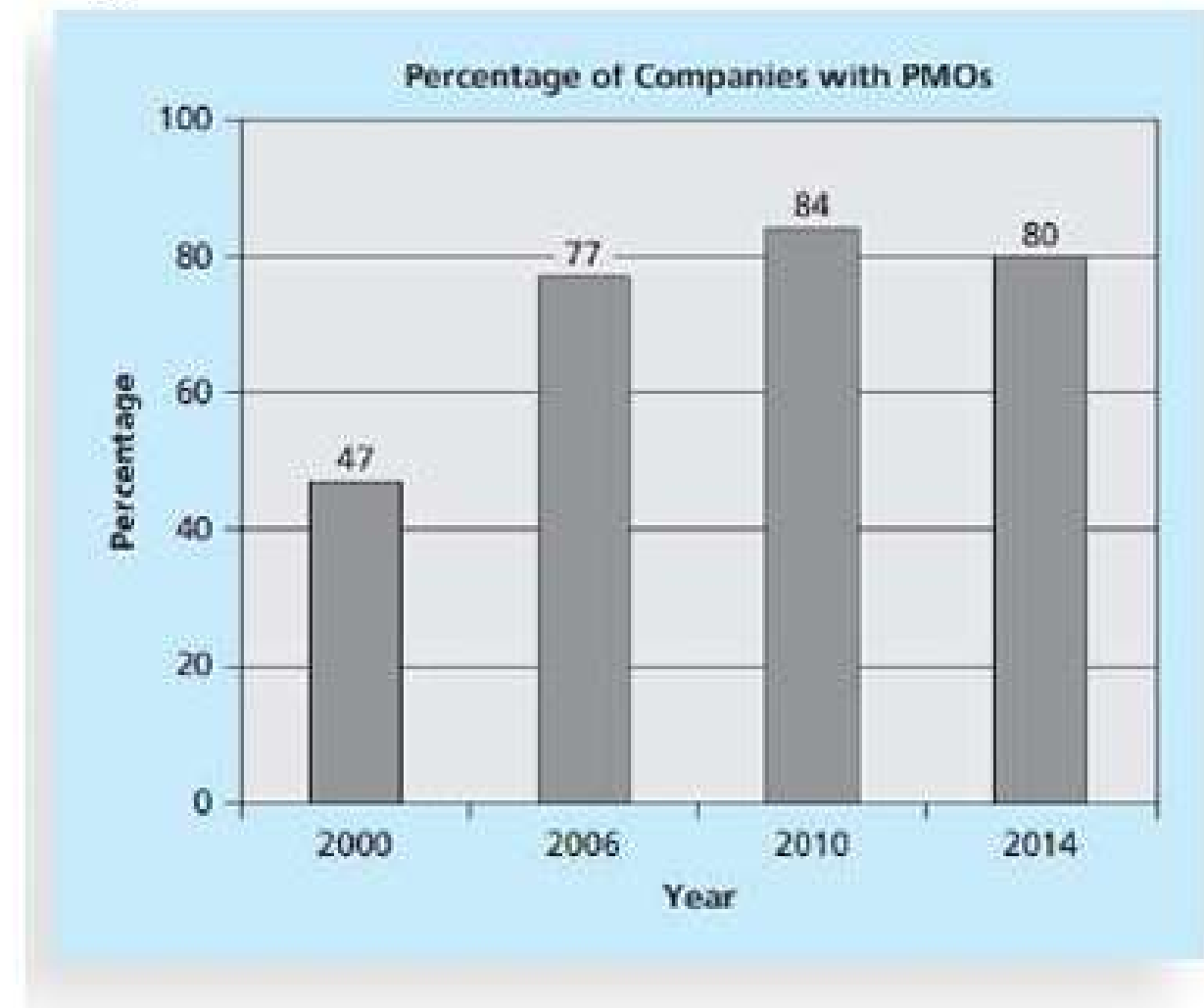
- Network Diagrams
- used to visually represent the sequence and dependencies of project activities.



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FIGURE 1.7 SAMPLE NETWORK DIAGRAM CREATED WITH PROJECT 13

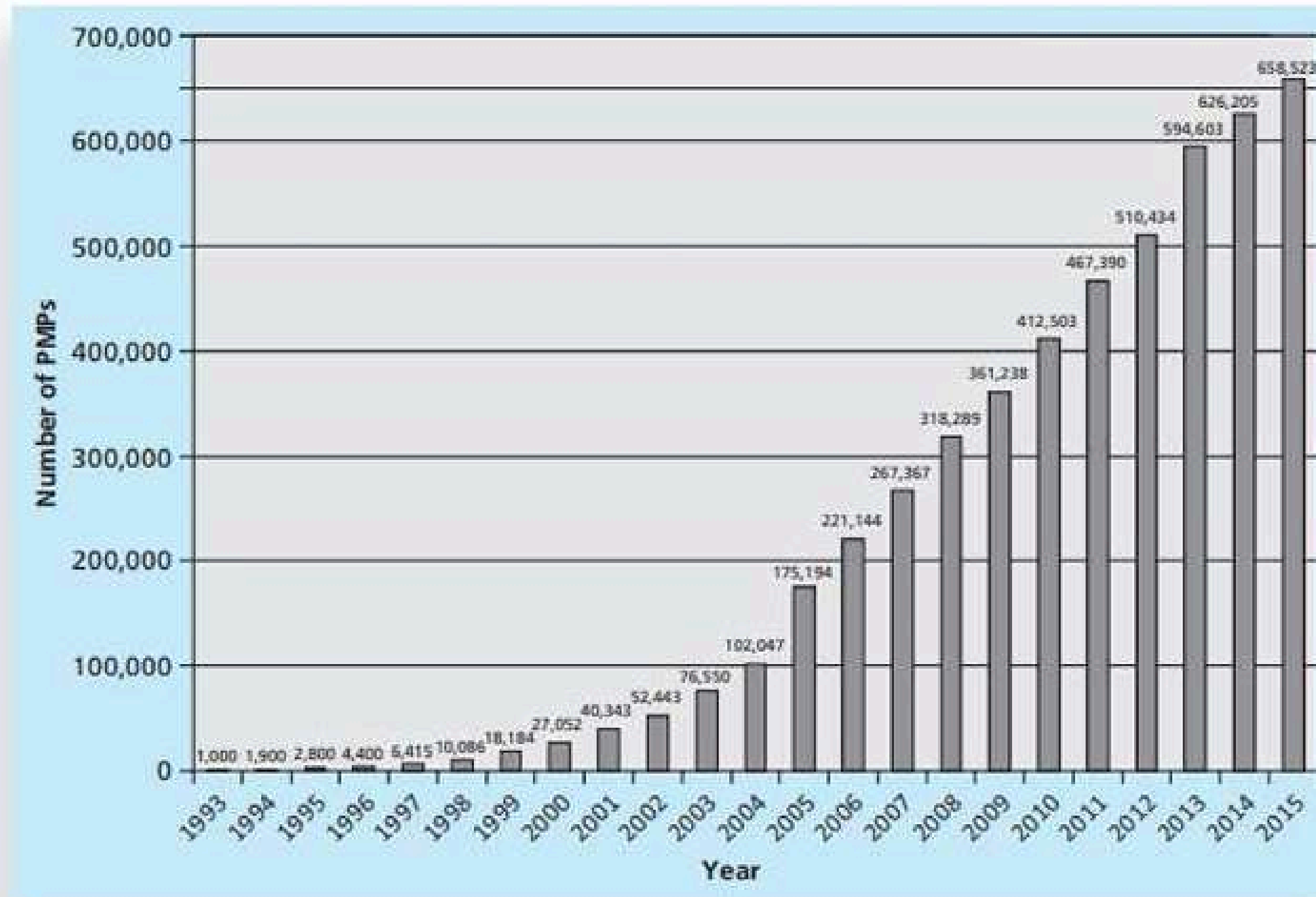
- 1990s, many companies began to create project management offices,
- PMO (Project Management Office)
- PMI (Project Management Institute)



Source: PM Solutions, "The State of the PMO 2010" (2010).

**FIGURE 1.8 GROWTH IN THE NUMBER OF PROJECT MANAGEMENT OFFICES**





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FIGURE 1.9 GROWTH IN PMP CERTIFICATION, 1993-2014



# Ethics in Project Management

is a set of principles that guides decision making based on personal values of what is considered right and wrong.

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# Conclusion:

There are many reasons to study project management, particularly as it relates to IT projects. The number of IT projects continues to grow in almost every industry the complexity of these projects continues to increase, and the profession of project management continues to expand and mature. As more people study and work in this important field, the success rate of IT projects should continue to improve.



The background of the slide is a blurred office scene. It features a laptop keyboard in the lower-left corner, a white calculator on a desk in the center-left, and various business charts and documents scattered around. A person's arm is visible in the bottom right, and a small potted plant is in the upper right. A dark blue semi-transparent rectangle is overlaid on the right side of the image, containing the text and decorative elements.

# Thank You

Presentation

